AI-Powered Interactive Dress Designer App

Shanto-Mariam University of Creative Technology

Project Name: AI-Powered Interactive Dress Designer App

Theme: Bridging Creativity & Manufacturing with AI

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Project Overview

- Core Concept: An innovative AI-based application for automatic generation, visualization, and customization of dresses.

- Key Features: Enables personalized styling by adjusting color, fabric, sleeves, necklines, and patterns.

- Objective: To make the design process engaging, efficient, and accessible for both individuals and professionals through real-time AI visualization.

Problem Statement

- Efficiency: Traditional design is manual, time-consuming, and expensive.
- Accessibility: Existing CAD tools (e.g., CLO 3D) are too complex for non-professionals.
- Sustainability: The industry struggles with overproduction and excessive material waste due to poor communication.
- Visual Gap: Lack of real-time tools for experimenting with new ideas before production.

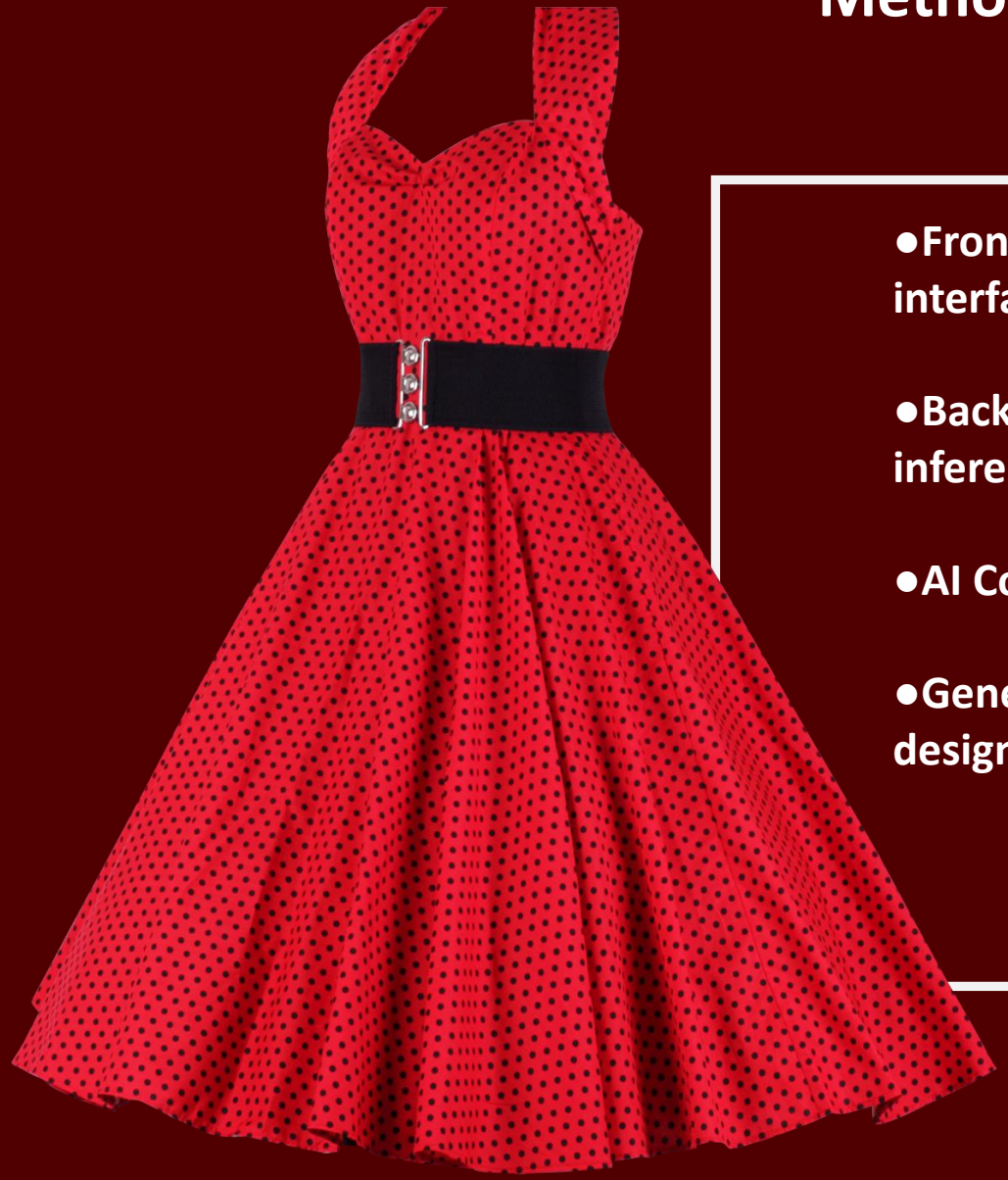
Research Gaps

- Interactive Granularity: Most AI models struggle with localized modifications (e.g., changing just a sleeve) without altering the whole dress.

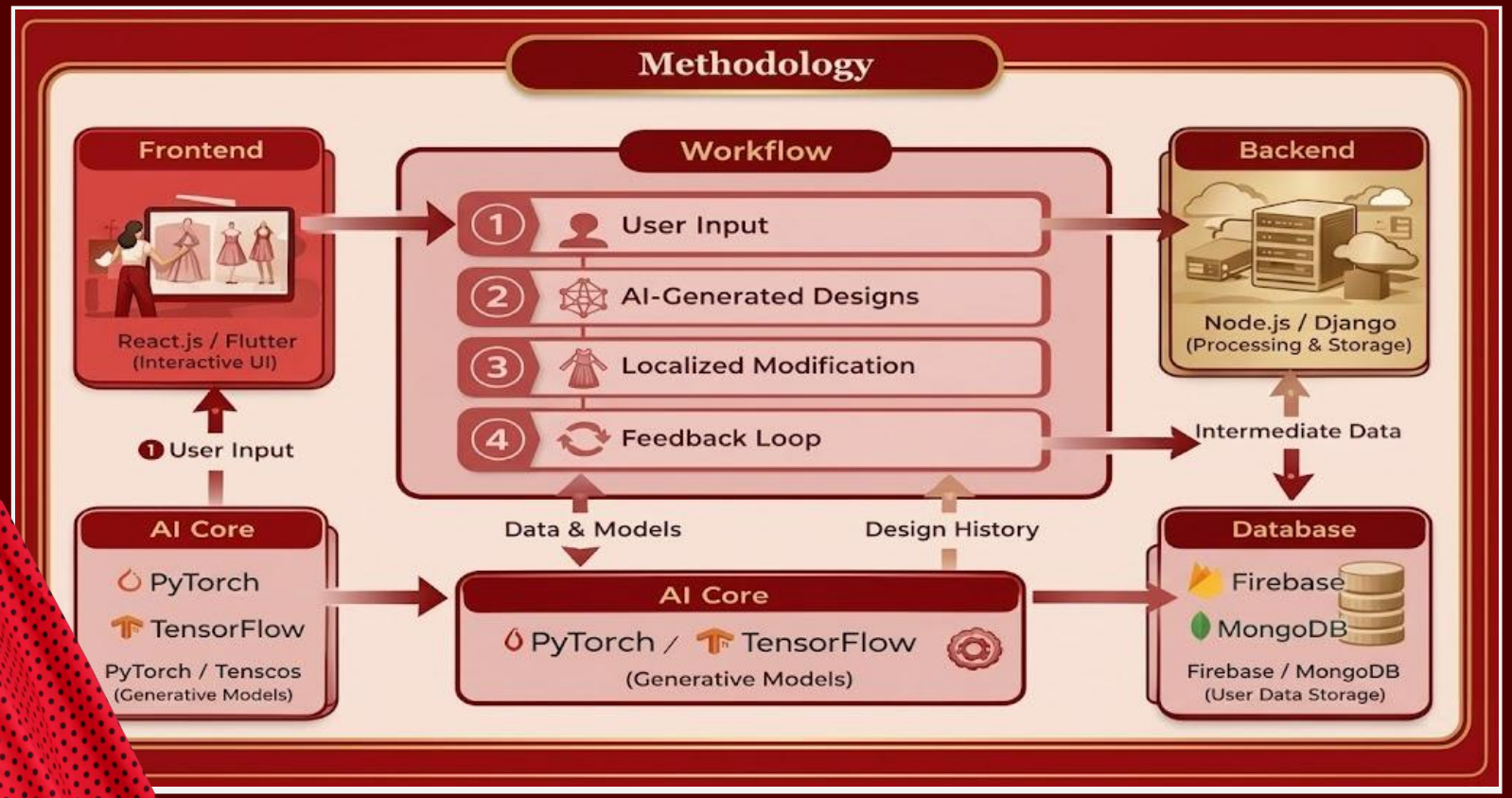
- Manufacturability: A "white space" exists between AI-generated images and structurally sound, sewable garments.

- Intelligent Guidance: Lack of AI assistants to guide users through fashion theory, such as harmonious fabric and color selection.

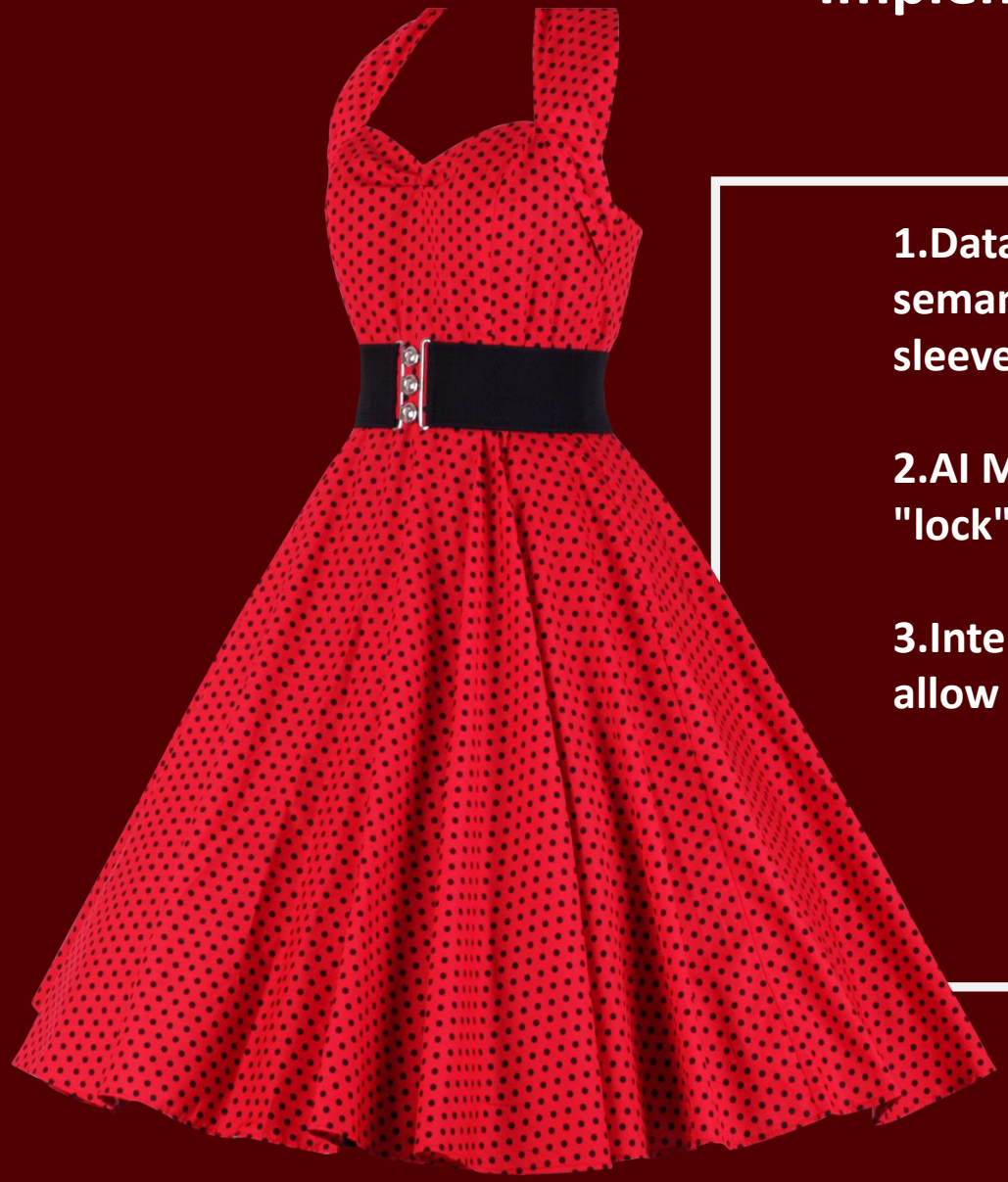
Methodology — System Architecture



- Frontend: Built with React.js or Flutter for a user-friendly interface.
- Backend: Powered by Node.js or Django to handle heavy AI inference and storage.
- AI Core: Utilizes PyTorch or TensorFlow frameworks.
- Generative Models: Employs Stable Diffusion or GANs for design creation.


Methodology
Frontend
React.js / Flutter
(Interactive UI)
1 User Input
AI Core
PyTorch
TensorFlow
PyTorch / Tenscos
(Generative Models)
Workflow
1 User Input
2 AI-Generated Designs
3 Localized Modification
4 Feedback Loop
Data & Models
Design History
AI Core
PyTorch / TensorFlow
(Generative Models)
Backend
Node.js / Django
(Processing & Storage)
Intermediate Data
Database
Firebase
MongoDB
Firebase / MongoDB
(User Data Storage)

Implementation Approach (Steps 1-3)



1.Data Acquisition: Curating datasets like DeepFashion, using semantic segmentation to define dress boundaries (neckline, sleeves, torso).

2.AI Model Development: Using Conditional GANs (cGAN) to "lock" specific attributes while modifying others.

3.Interactive Layer: Implementing Latent Vector modifications to allow real-time "Inpainting" when a user moves a slider.

Implementation Approach (Steps 4-6)

4.Backend & Scalability: Utilizing model quantization to ensure AI predictions run on cloud GPUs in under 2 seconds.

5.Testing & Validation: Evaluating designs based on "Aesthetic Accuracy" and "Structural Feasibility" (can it actually be sewn?).

6.Database: Managing user history and design versions via Firebase or MongoDB.

User Experience Flowchart

- Initialization: Starts with 3D face scanning for personalized modeling.
- Design Loop: Users load clothing models and iterate until the style is satisfactory.
- Customization: Parallel paths for "Personalized design" (fabrics/buttons) and "Interactive appearance" (rotate/scale).
- Finalization: Satisfaction check leads directly to placing an order.

Conclusion & Future Impact

- Digital Innovation: Simplifies the creative process through smart, accessible tools.
- Sustainability: Promotes material efficiency by visualizing designs before they are finalized.
- Production-Ready: Transitions from "visually stunning" images to downloadable production files.

THANK YOU